Machine learning: introduction

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Outline

1. AI, machine learning and related fields
2. Machine learning – overview, definitions
3. Branches of machine learning
4. Nearest neighbour classification
5. Data scaling

Artifical intelligence - AI

- AI is „hot topic“… and you can see or hear this term almost everywhere.
- Actually, people have started using the term when they refer to applications, that used to be called by other names.

Why?

- There is no „official“ definition; the AI field is still developing and methods, which were considered as belonging to AI fifty years ago are now taught as a part of statistical course or computer science.
- Because the problems that looks easy is difficult to teach some robot or computer.
- And problems that looks difficult is easy to teach some robots or computer.

How would you describe key features of AI?

- Autonomy
  The ability to perform tasks in complex environments without constant guidance by a user.

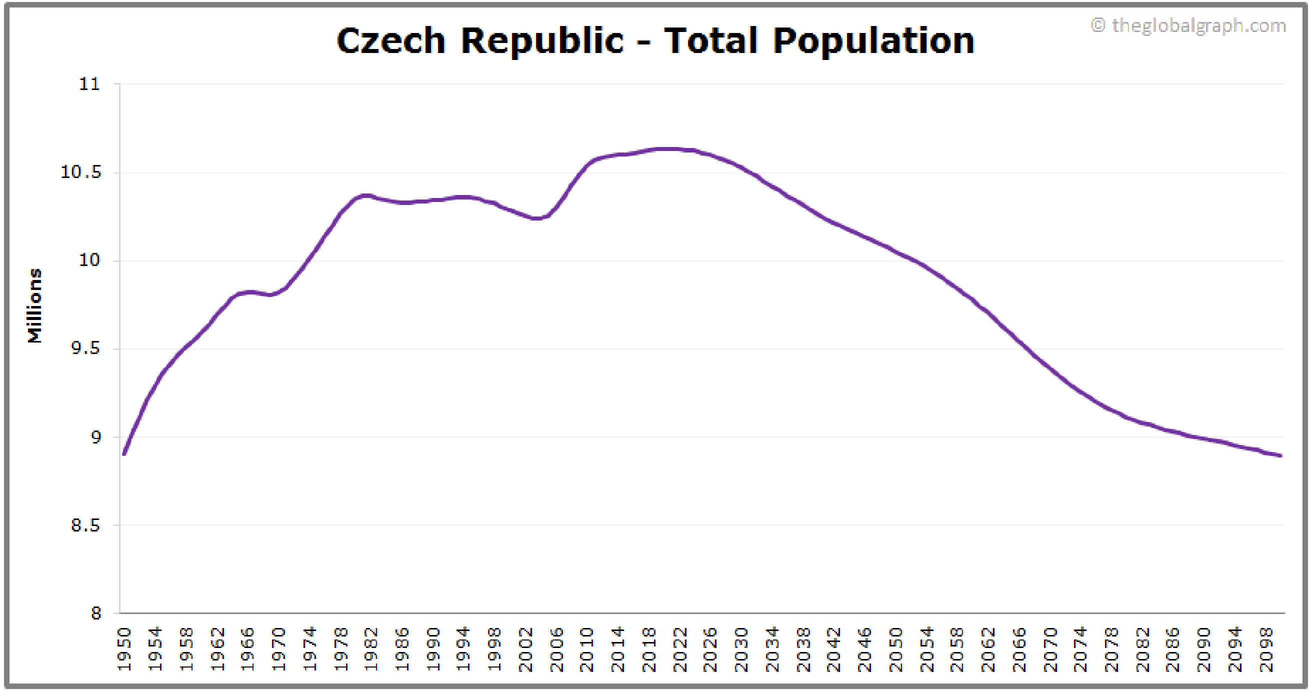
- Adaptivity
  The ability to improve performance by learning from experience.

However, when talking about the AI, we must be careful, as many of the words that we use can be quite misleading.
Common examples: *learning, understanding,* and *intelligence*.

Is this application of AI?

No

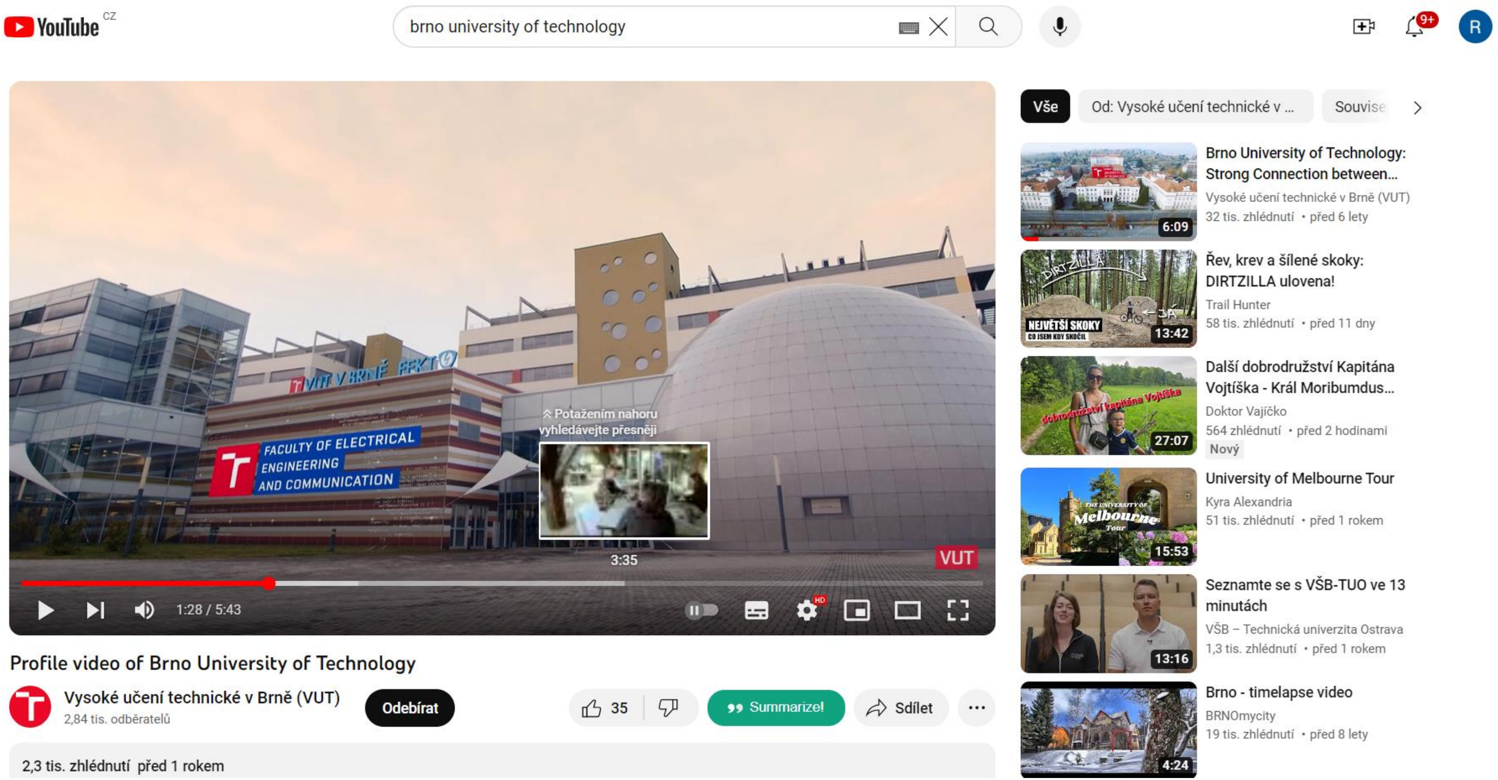
Czech republic - population prediction.



Is this application of AI?

Kind of

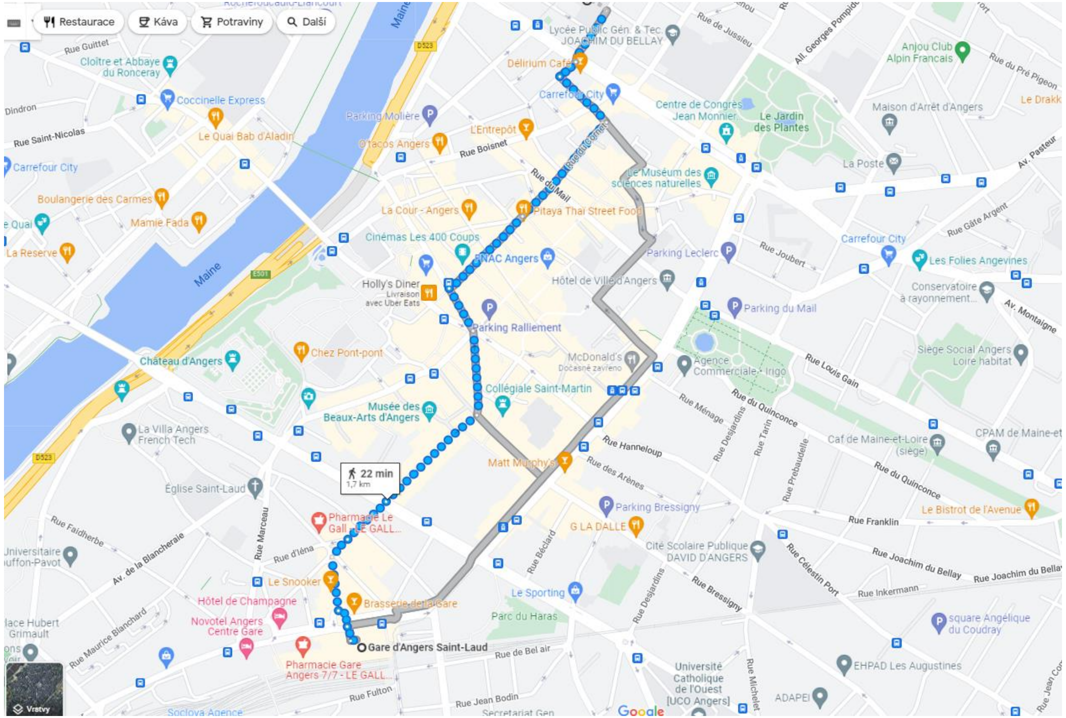
A video recommendation system such as Youtube that suggests video based on the users' watching behavior.



Is this application of AI?

No

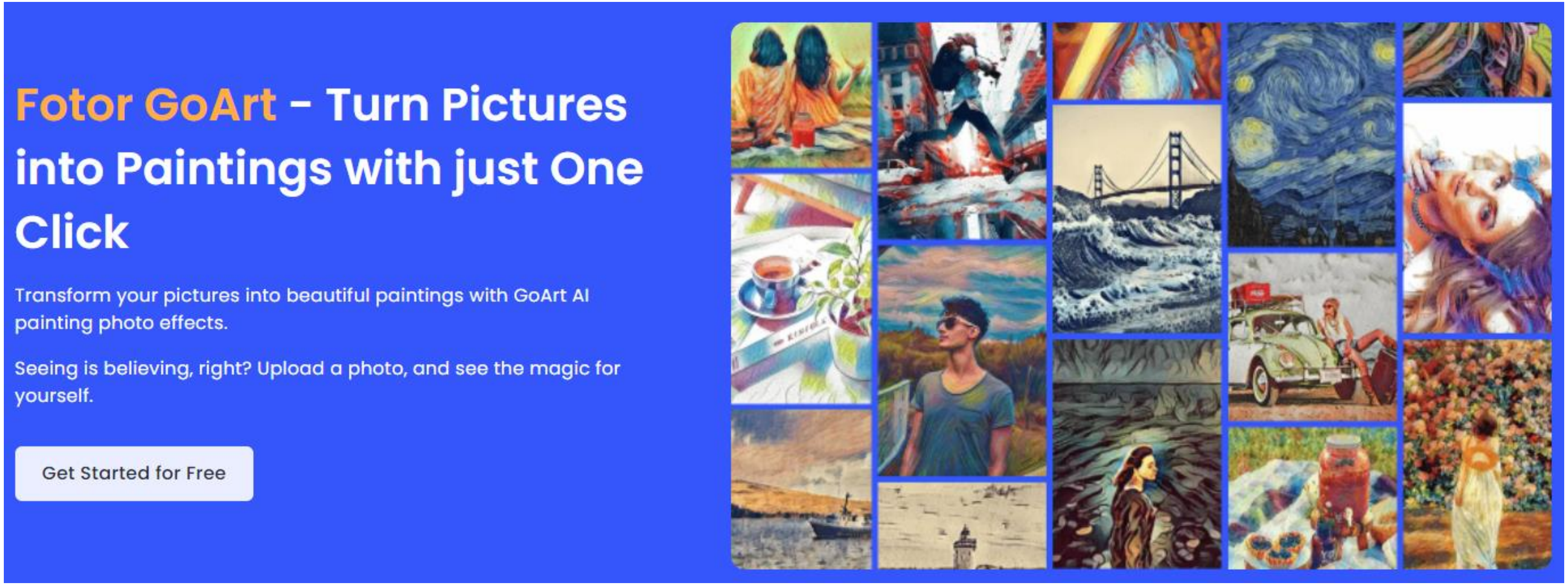
- A GPS navigation system for finding the fastest route



Is this application of AI?

Kind of

- Style transfer filters in applications such as Prisma that take a photo and transform it into different art styles (impressionist, cubist, ...)



See nice demo here:
https://www.linkedin.com/feed/update/urn:li:activity:6926842024608337920?utm_source=linkedin_share&utm_medium=member_desktop_web

Related topics

- In addition to AI, there are several other closely related topics that are good to know at least by name. These include mainly:
  - machine learning
  - computer vision
  - robotics
  - data science
  - deep learning

Machine learning

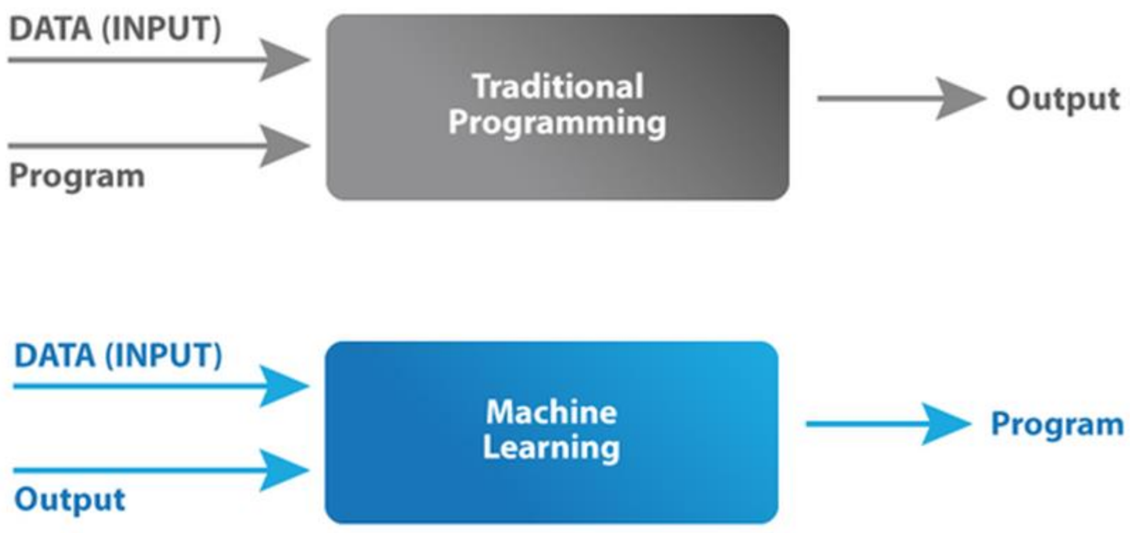
- Systems that improve their performance in a given task with more and more experience or data.

- Machine learning can be said to be a subfield of AI, which itself is a subfield of computer science (such categories are often somewhat imprecise and some parts of machine learning could be equally well or better belong to statistics).

- Machine learning enables AI solutions that are adaptive, i.e. machine learning enables adaptivity.

Machine learning

The term Machine Learning was coined by Arthur Samuel in 1959, an American pioneer in the field of computer gaming and artificial intelligence and stated that “it gives computers the ability to learn without being explicitly programmed”.



https://d1m75rqqgidzqn.cloudfront.net/2019/10/What-is-Machine-Learning-Machine-learning-model-vs-traditional-model.jpg

Recent ML definitions

“Machine Learning at its most basic is the practice of using algorithms to ***parse data****,* ***learn from it****, and then* ***make a determination or prediction*** *about something in the world.”* **–** *Nvidia*

“Machine Learning is the science of ***getting computers to learn*** *as well as humans do or better.” - Dr. Roman Yampolskiy, University of Louisville*

“Machine Learning is the science of ***getting computers to learn*** *and* ***act*** *like humans do, and* ***improve*** *their learning over time in* ***autonomous fashion****, by feeding them data and information in the form of observations and real-world interactions.” -* [*https://emerj.com/ai-glossary-terms/what-is-machine-learning/*](https://emerj.com/ai-glossary-terms/what-is-machine-learning/)

„Machine learning is a branch of AI and computer science which focuses on the **use of data and algorithms** to imitate the way that humans **learn**, gradually **improving** its accuracy.“ - *IBM*

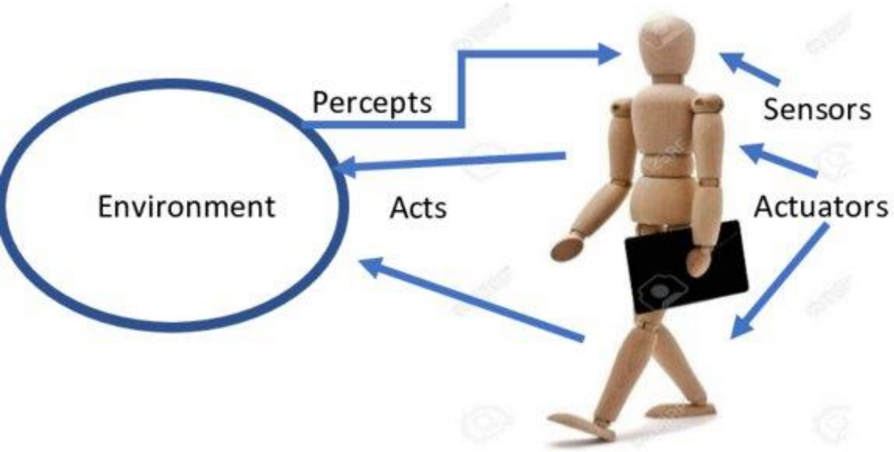
Computer vision

- Computer vision is a field of artificial intelligence (AI) that enables computers and systems to derive meaningful information from digital images, videos and other visual inputs — and take actions or make recommendations based on that information.

https://www.ibm.com/topics/computer-vision

Robotics

- Robotics means building and programming robots so that they can operate in complex, real-world scenarios.
  - Robotics requires combination of virtually all areas of AI, mainly:
    - Computer vision, speech recognition and other
    - Natural language processing, information retrieval, and reasoning under uncertainty for processing instructions and predicting consequences of potential actions
    - Cognitive modeling and affective computing, i.e. systems that respond to human actions, or human feelings, for interacting and working together with humans



https://www.researchgate.net/publication/357618741_Challenges_of_Artificial_Intelligence_--_From_Machine_Learning_and_Computer_Vision_to_Emotional_Intelligence

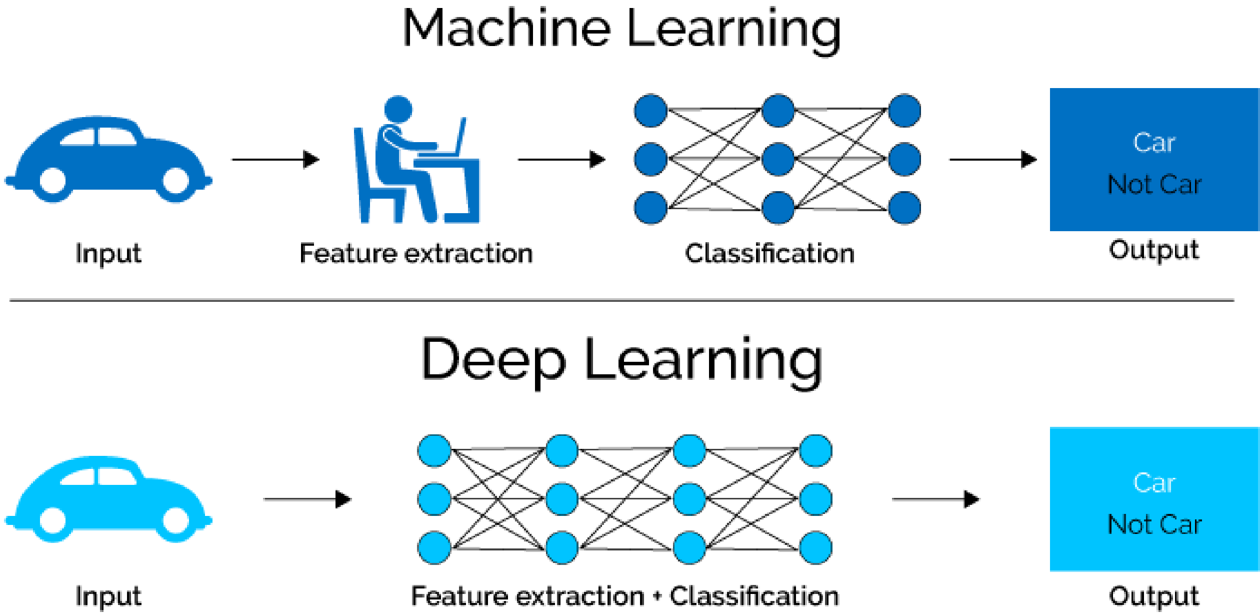
Data science

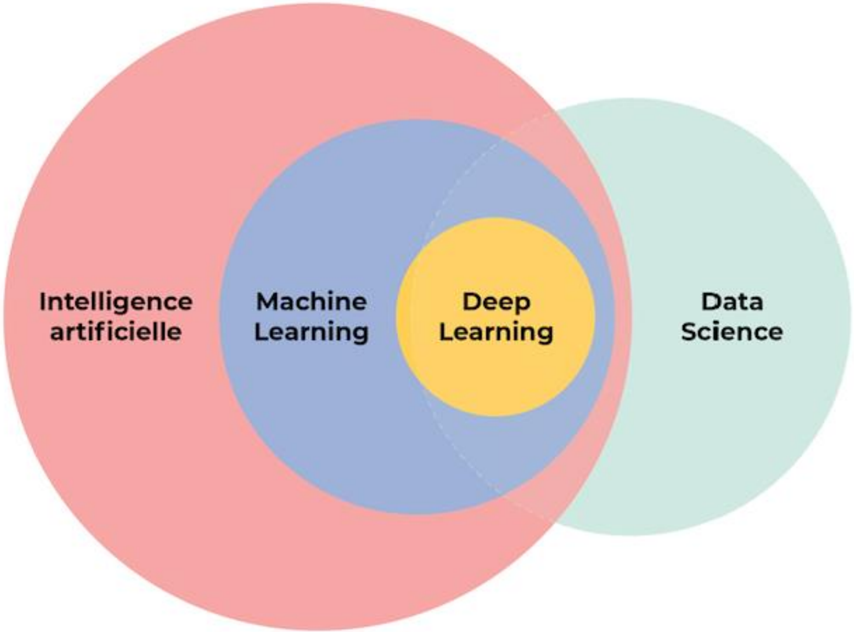
- Data science covers several subdisciplines that includes **machine learning**, **statistics**, and **computer science** (some specific algorithms, data storage, web application development).

- Data science is typically a practical discipline and it usually requires understanding of the domain in which it is applied in (e.g. business, trade market, biology…).

Deep learning

- Deep learning is a subfield of machine learning.
- „Deep“ refers to the complexity of a mathematical model, e.g. some classifiers.





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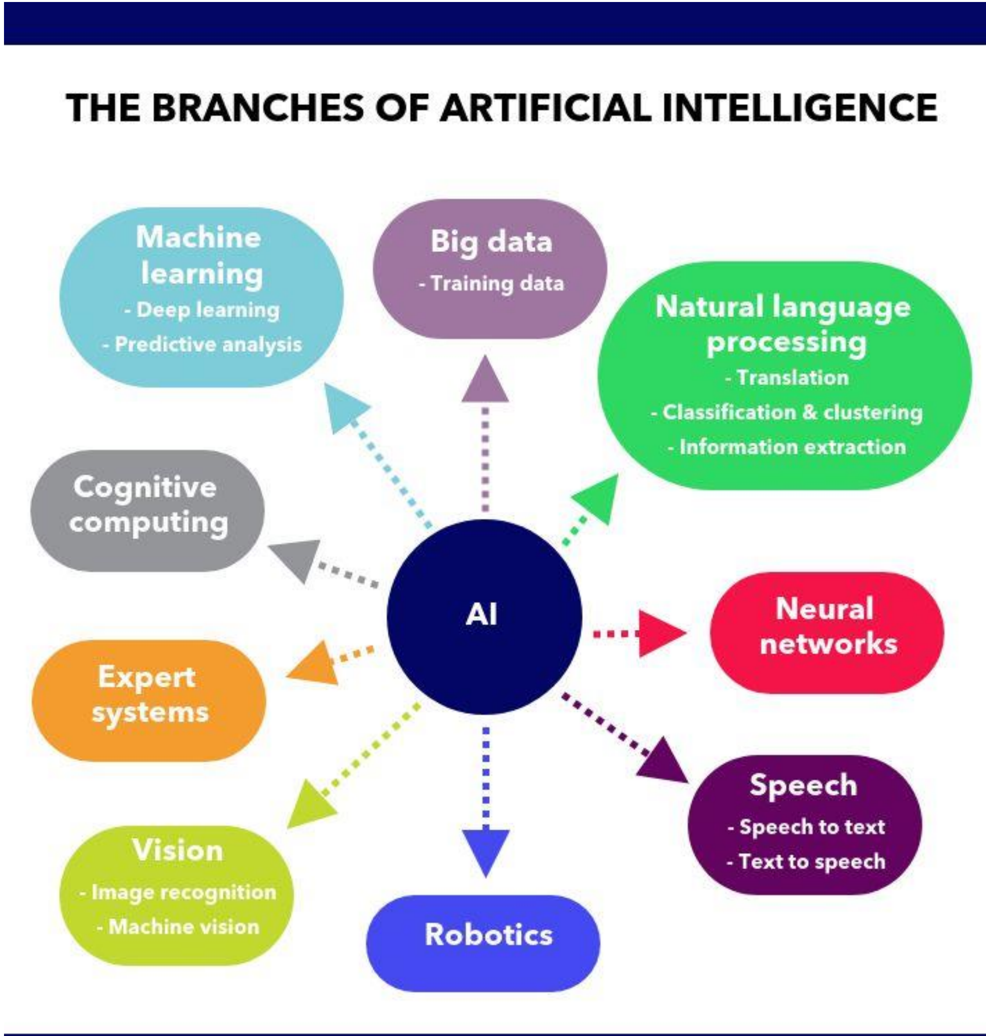


https://www.pinterest.com/pin/698762317082278287/

AI – a topic for philosophers

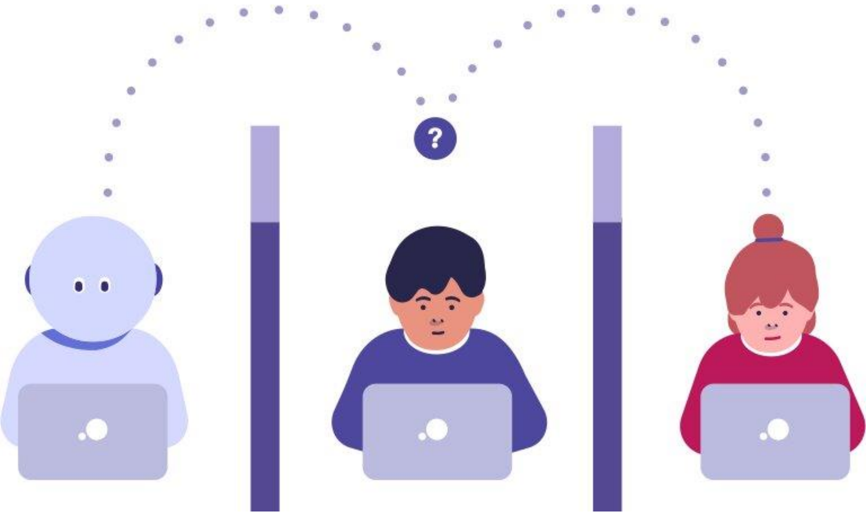
AI brings up philosophical questions whether intelligent behavior implies or requires the existence of a mind…

If a machine behaves in an intelligent manner… does it means that it is intelligent? Or does it means that it has a “mind” in the way that a human has?

Turing test

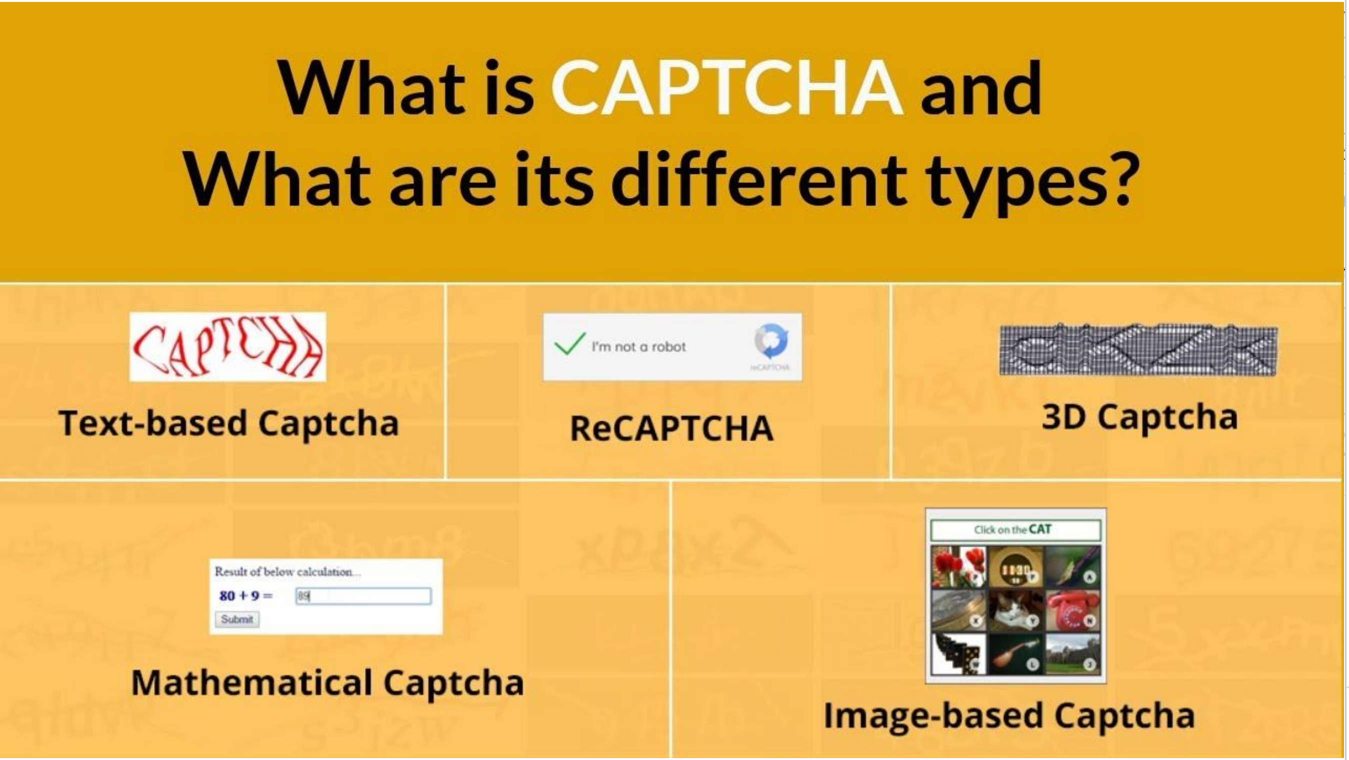
In the test, a human interrogator interacts with two players, A and B, in a chat. If the interrogator cannot determine which player, A or B, is a computer and which is a human, the computer is said to pass the test.

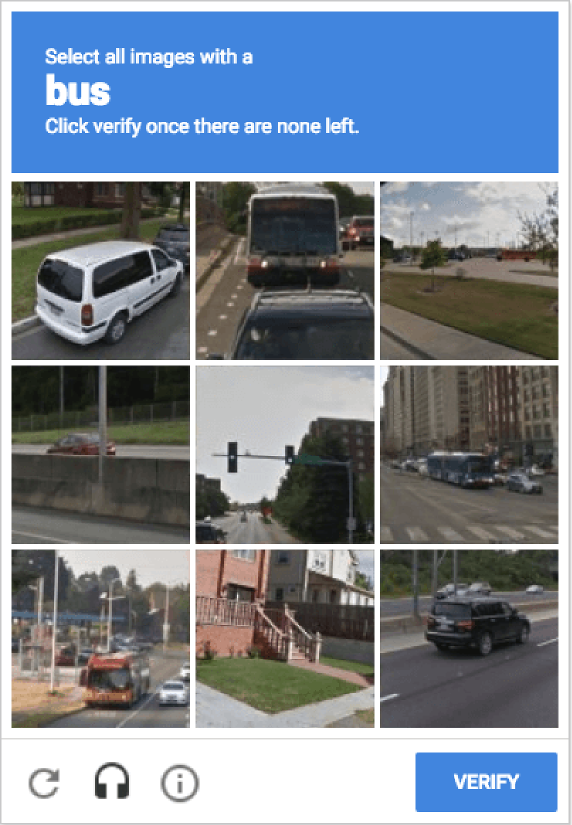
The argument is that if a computer is indistinguishable from a human in a general natural language conversation, then it must have reached human-level intelligence.



https://visiongenius.ai/blog/what-is-the-turing-test

Captcha


What is CAPTCHA and
What are its different types?
CAPTCHA
Text-based Captcha
I'm not a robot
ReCAPTCHA
3D Captcha
Result of below calculation...
80 + 9 =
89
Submit
Mathematical Captcha
Click on the CAT
Image-based Captcha


Select all images with a
bus
Click verify once there are none left.
VERIFY

Is a self-driving car an “intelligent car” ?

A self-driving car is an example of an element of intelligence (driving a car) … but driving the car can be automated.

The AI system in the car doesn’t see or understand its environment, and it doesn’t know how to drive safely, in the way a human being sees, understands, and knows.

The intelligent behavior of the system is fundamentally different from actually being intelligent.

Terminology: general vs narrow AI

Narrow AI refers to AI that handles one task.

General AI, or Artificial General Intelligence (AGI) refers to a machine that can handle any intellectual task.

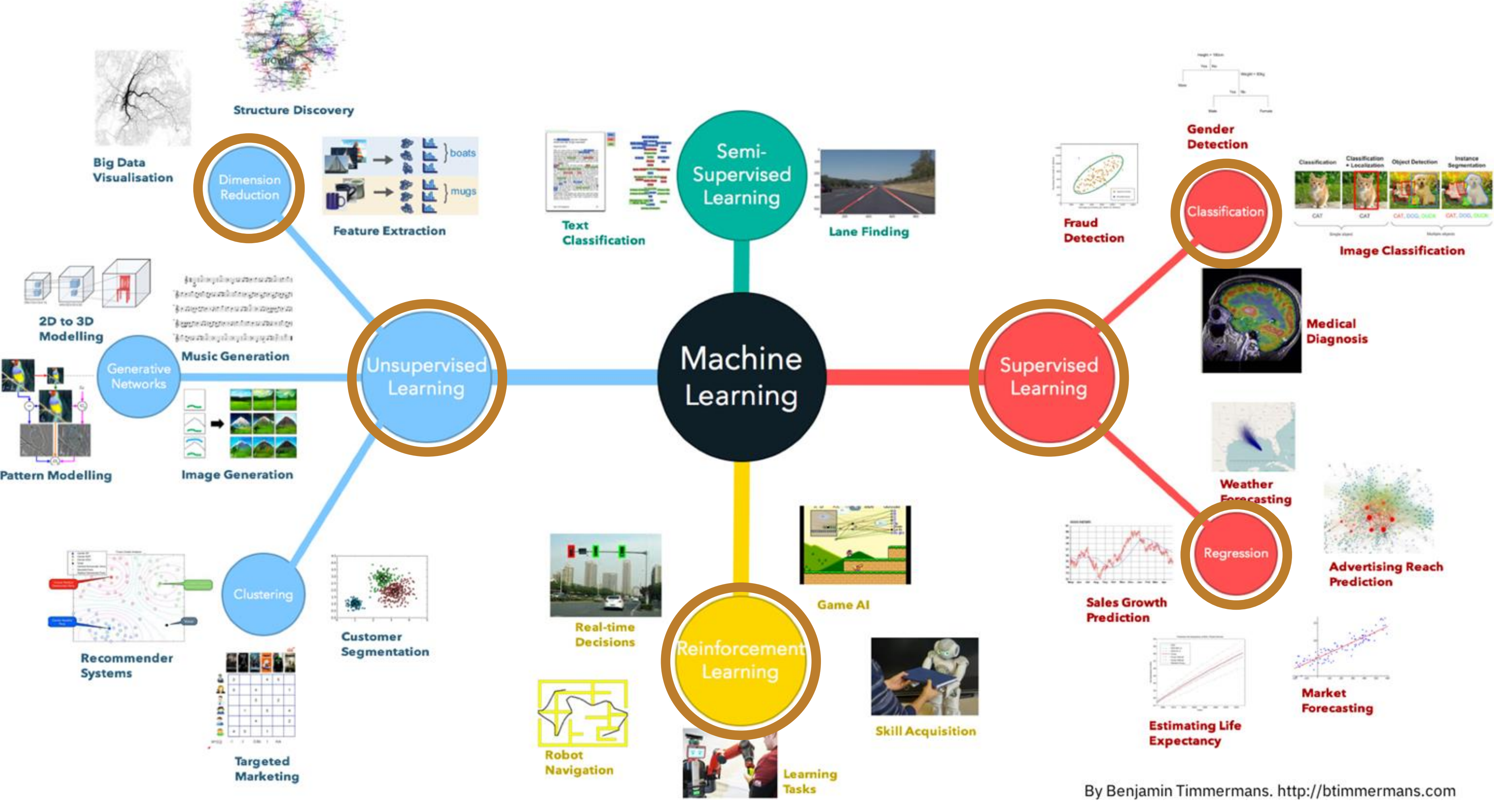
All the AI methods we use today fall under narrow AI, with general AI being in the realm of science fiction.

Terminology: strong vs weak AI

Strong AI point to a “mind” that is genuinely intelligent and self-conscious.

Weak AI is what we actually have - namely systems that exhibit intelligent behaviors despite still being “only“ computers.

Machine learning

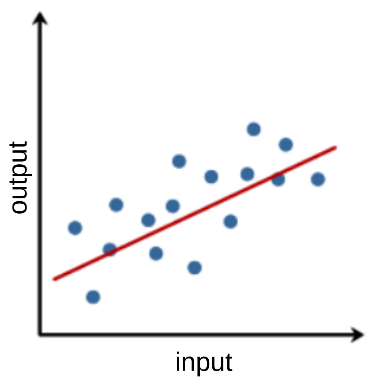

Machine Learning
Unsupervised Learning
Semi-Supervised Learning
Supervised Learning
Reinforcement Learning
Dimension Reduction
Big Data Visualisation
Structure Discovery
Feature Extraction
boats
mugs
Generative Networks
2D to 3D Modelling
Music Generation
Pattern Modelling
Image Generation
Clustering
Recommender Systems
Targeted Marketing
Customer Segmentation
Text Classification
Lane Finding
Classification
Gender Detection
Fraud Detection
Image Classification
Medical Diagnosis
Regression
Weather Forecasting
Advertising Reach Prediction
Sales Growth Prediction
Estimating Life Expectancy
Market Forecasting
Real-time Decisions
Game AI
Robot Navigation
Learning Tasks
Skill Acquisition
By Benjamin Timmermans. http://btimmermans.com

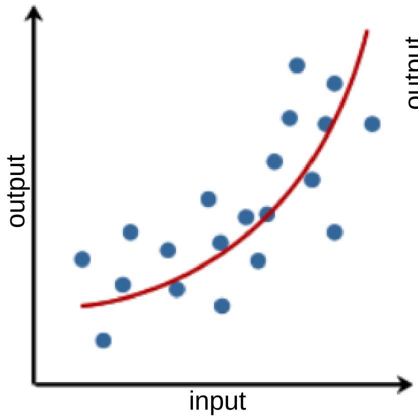
Regression

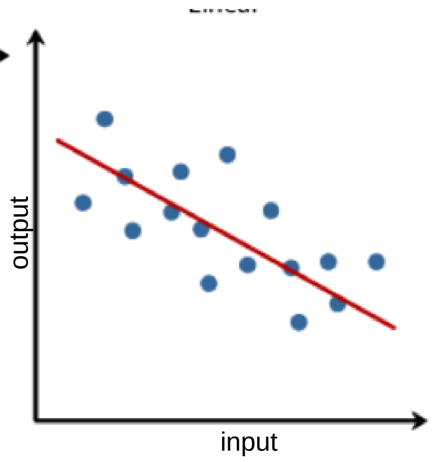
x - Input – independent variable, feature
y - Output – dependent variable, target

Regression - the main output of the model is a function *f* that automatically produces a prediction, denoted with $\hat{y}$, for a vector of predictors (i.e. feature samples): $\hat{y} = f(\mathbf{x})$
So we want this prediction to match the actual outcome as best as possible. Because the outcome is continuous, our predictions will not be either exactly right or wrong, but instead we will determine an error defined as the difference between the prediction and the actual outcome:

$$|\hat{y} - y|$$



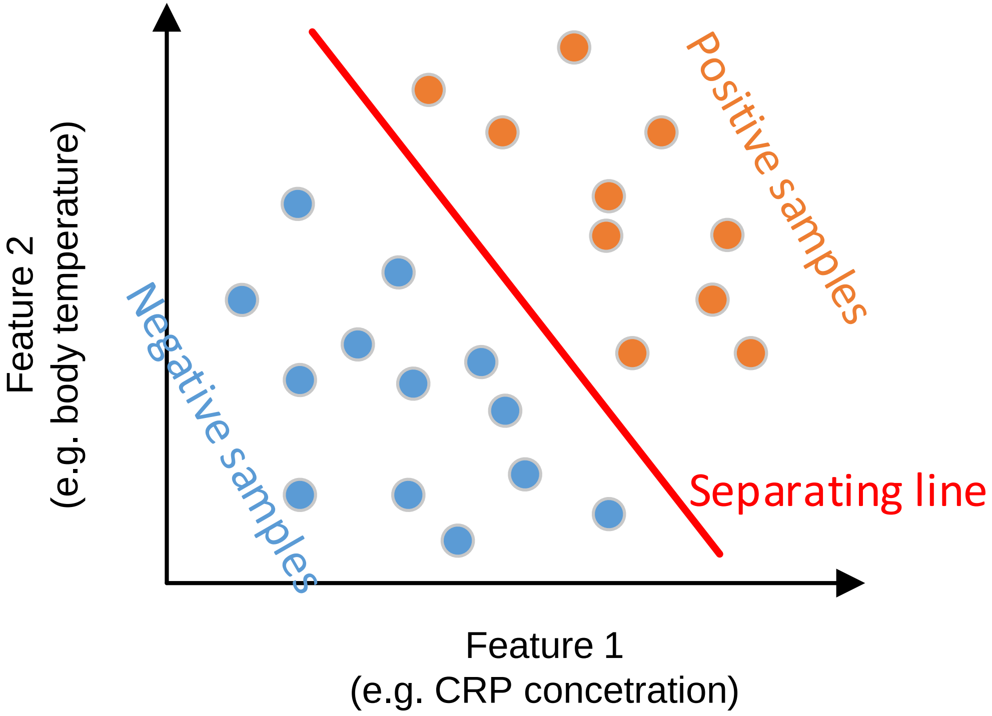




https://statistics.laerd.com/spss-tutorials/

Classification

In classification the main output of the model will be a decision rule which prescribes which of the *K* classes (i.e. discrete values; there can be from *2* to *N* classes) we should predict based on input feature vector $\mathbf{x}$. In this scenario, most models provide functions of the predictors for each class $f_k(\mathbf{x})$, that are used to make this decision.

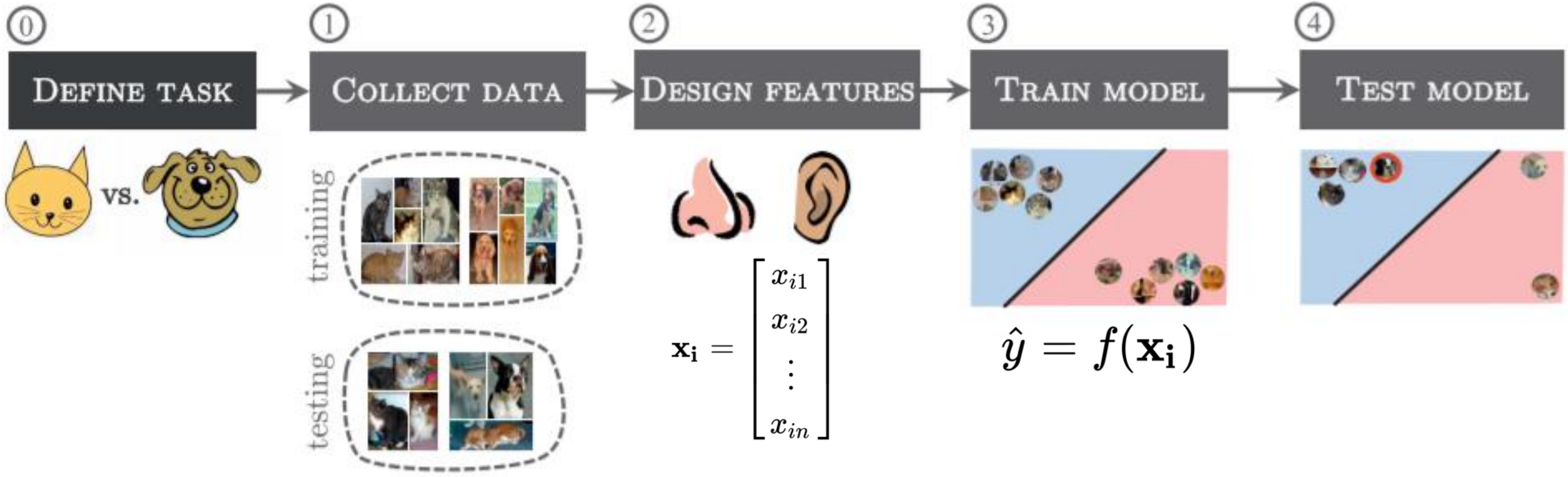


What is „feature“ ?

A FEATURE IS AN INDIVIDUAL MEASURABLE PROPERTY OR CHARACTERISTIC OF A PHENOMENON.

Classification

A typical (now, we can also say traditional) machine learning pipeline can be split into several steps.



Wat, Borhani, Katsagfelos: Machine learning refined, Cambridge University Press, 2019

The typical classification pipeline

0. **Define the problem** - what is the task we want to teach a computer to do? Detection of objects in images? Classification? Regression?

1. **Collect the data** for training and evaluation - the larger and more diverse the data the better.

2. **Design the features** - what features describe best our data? Decide if some feature selection or reduction is needed. The features then forms a feature vector $\mathbf{x_i}$ for one *measurement* (with possible label y_i), for i=1,...n.

3. **Select, set and train the model** - this model takes the feature vector and gives an outcome (output values): $\hat{y} = f(\mathbf{x_i})$

4. **Test the model** - evaluate the performance of the trained model. Tune the parameters of the model.

Multiclass and multilabel classification

- **multiclass** (K > 2); e.g. classification of objects in the image (car, bus, train, bike...)

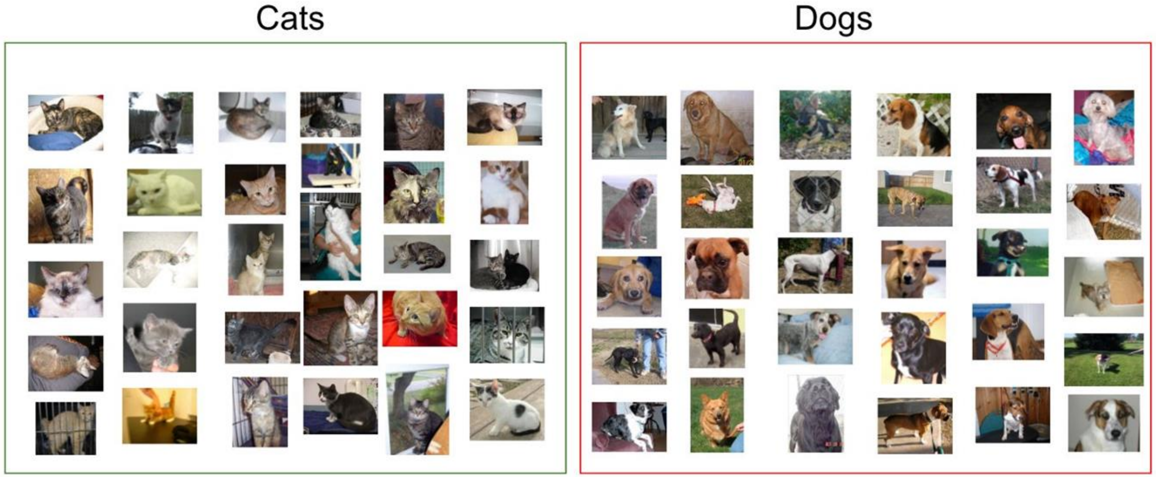



- **multilabel** - one input can belong to more classes;
e.g.the image can contain different objects (car and bike)



Supervised learning

The machine has a "supervisor" or a "teacher" who gives the machine all the answers, like whether it's a cat in the picture or a dog. The “teacher” has already divided (labeled) the data into cats and dogs, and the machine is using these examples to learn.



Sample of cats & dogs images from Kaggle Dataset

Supervised learning

Data: $(\mathbf{x_i}, y_i)$

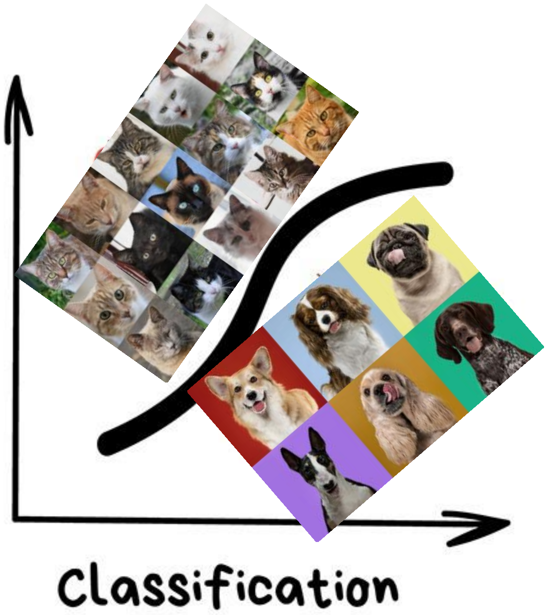
x is data, usually in the form of vectors of features i.e.

$$\mathbf{x_i} = \begin{bmatrix} x_{i1} \\ x_{i2} \\ . \\ . \\ x_{in} \end{bmatrix}$$

y represents labels of classes, for example:

y_i = cats (mathematically denoted as +1)

y_i = dogs (mathematically denoted as -1)



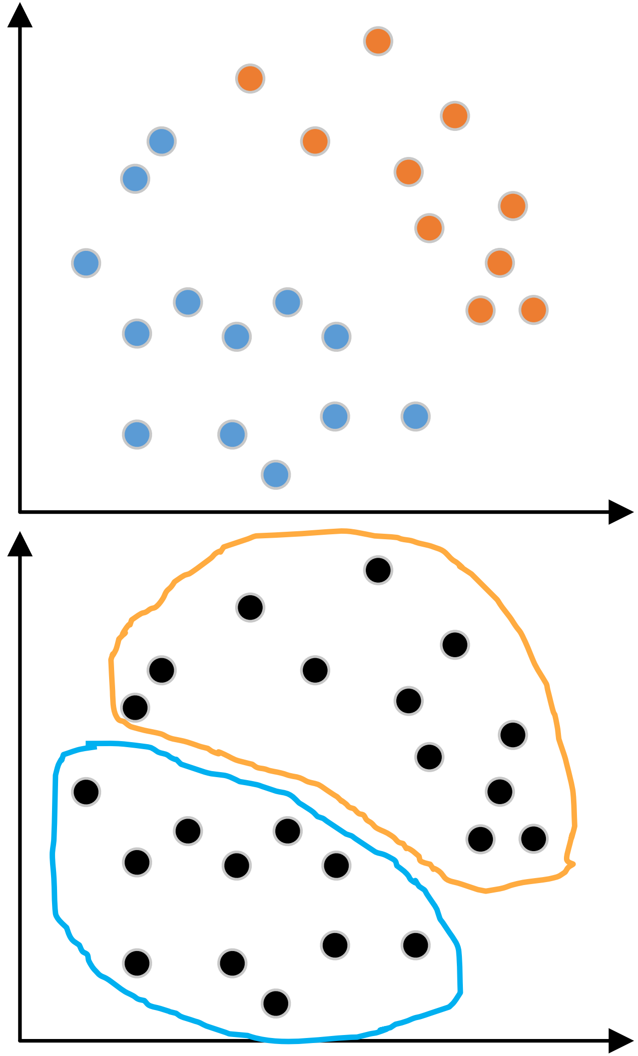
Goal: Learn a function to map x to y

Unsupervised learning

Data: Just data, no labels.

Goal: Learn some *hidden* structure of the data.

Example: Clustering.

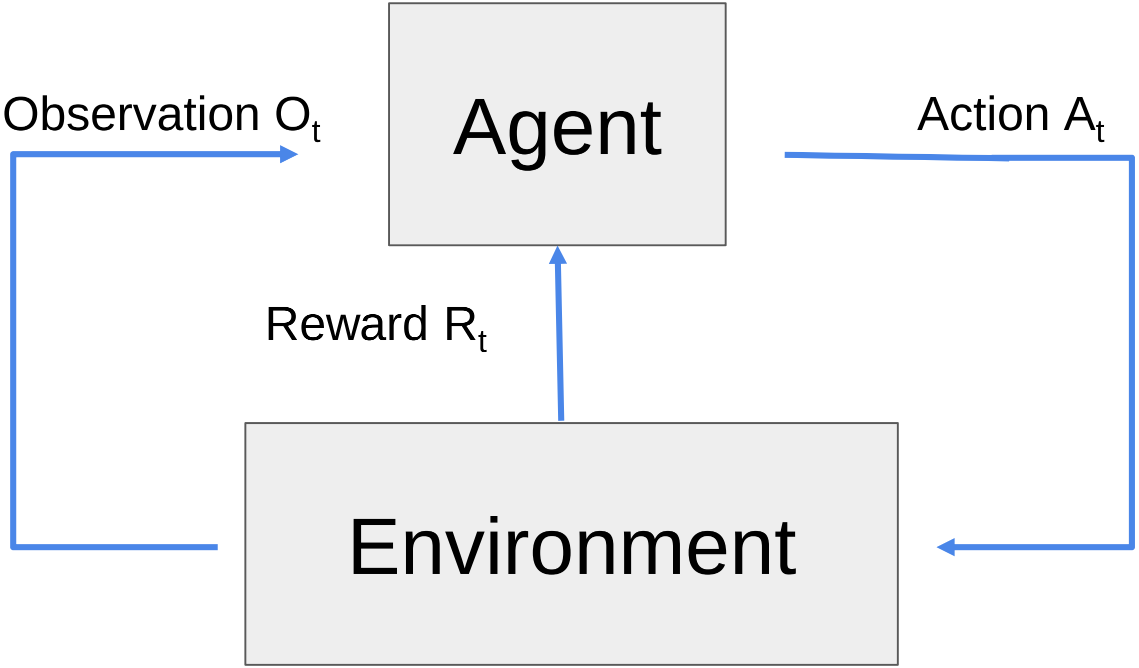


Reinforcement learning

Reinforced learning uses a straightforward approach - the action takes place, the consequences are observed, and the next action considers the results of the first action.

So, it is about taking suitable **action** to maximize **reward** in a particular situation. It is employed by various software and machines to find the best possible behavior or path it should take in a specific situation. Reinforcement learning differs from the supervised learning in a way that in supervised learning the training data has the answer key with it so the model is trained with the correct answer itself whereas in reinforcement learning, there is no answer but the reinforcement agent decides what to do to perform the given task.

Reinforcement learning



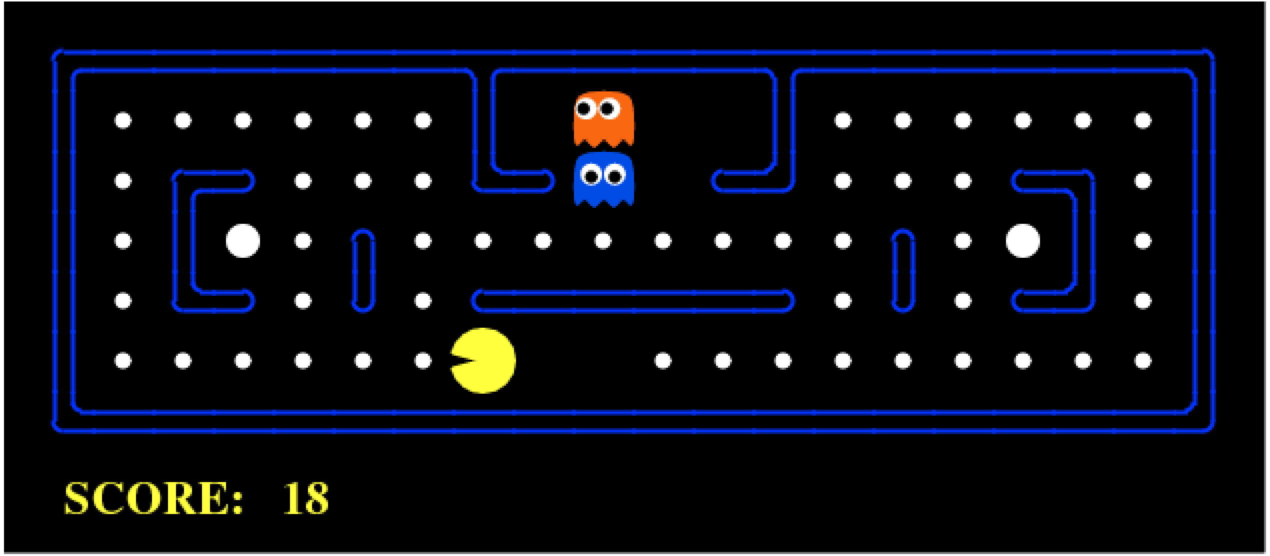
At each step the agent:
- executes action A_t
- receives observation O_t
- receives scalar rewards R_t

The environment:
- receives action A_t
- emits observation O_t
- emits scalar rewards R_t

Goal: select actions to maximize total future reward

Remarks: Actions may have long term consequences; Reward may be delayed; It may be better to sacrifice immediate reward to gain more long-term reward.

RL with PacMan



PacMan game - the goal of the **agent** (PacMan) is to eat the food in the grid while avoiding the ghosts on its way. In this case, the grid world is the **interactive environment** for the agent where it acts. Agent receives a **reward** for eating food and **punishment** if it gets killed by the ghost (loses the game). The states are the location of the agent in the grid world and the total cumulative reward is the agent winning the game.

Reinforcement learning

Examples:

PacMan:

https://youtu.be/QilHGSYbjDQ

Chicken:

https://www.youtube.com/watch?v=spfpBrBjntg

How DeepMind Conquered Go With Deep Learning (AlphaGo):
https://www.youtube.com/watch?v=IFmj5M5Q5jg

Self-driving car:
https://www.youtube.com/watch?time_continue=129&v=eRwTbRtnT1I

Stock prediction:
https://www.youtube.com/watch?v=05NqKJ0v7EE

The nearest neighbor classifier

- The nearest neighbor classifier is among the simplest possible classifiers. When given an item to classify, it finds the training data item that is most similar to the new item, and outputs its label.

What do we mean by nearest?

- The previous image shows the principle by simple visualization in 2-D space, i.e. for two features.
- Of course, there can be much more features, n-D space.

- The nearest neighbor classifier uses some specific definition of distance or similarity between instances.

- In the illustration above, we assumed that the *standard* geometric distance, technically called the Euclidean distance, is used.

What do we mean by nearest?

- Using the geometric distance to decide which is the nearest item may not always be reasonable or even possible: the type of the input may, for example, be text, where it is not clear how the items are drawn in a geometric representation and how distances should be measured.

- You should therefore choose the distance metric on a case-by-case basis.

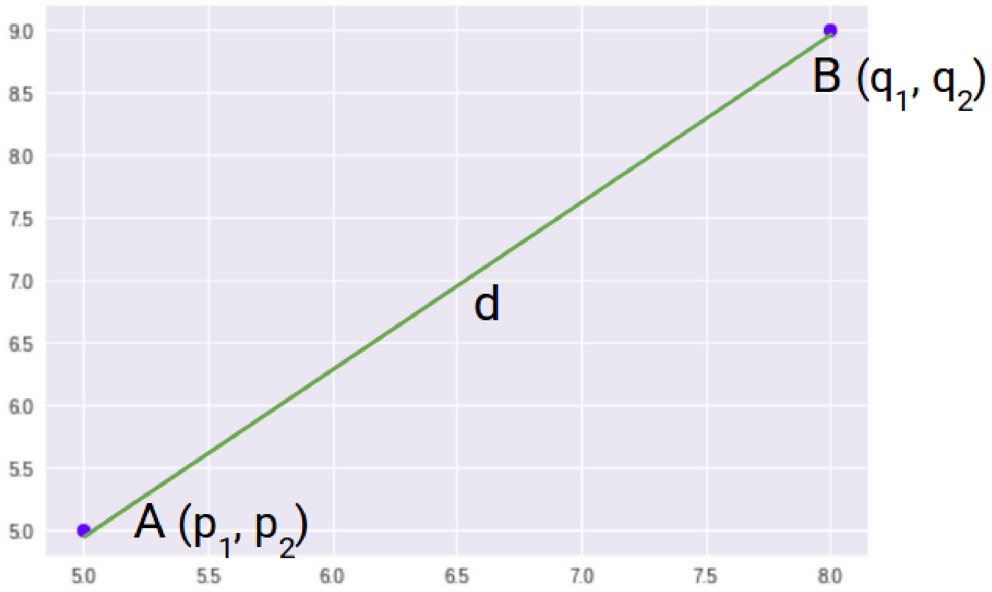
Euclidean distance

For 2D the distance is simply:

$$d = ((p_1 - q_1)^2 + (p_2 - q_2)^2)^{1/2}$$

We can generalize this for an n-dimensional space as:

$$D_e = \left(\sum_{i=1}^{n} (p_i - q_i)^2 \right)^{1/2}$$



https://www.analyticsvidhya.com/blog/2020/02/4-types-of-distance-metrics-in-machine-learning/

Manhatan distance

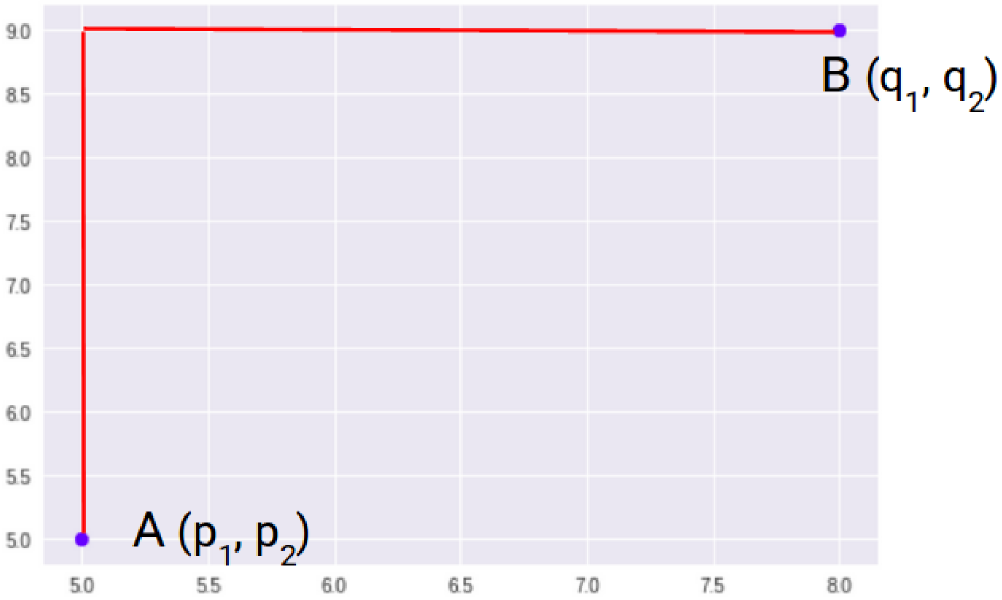
Manhattan Distance is the sum of absolute differences between points across all the dimensions.

Thus for 2D is it simply:

$$d = |p_1 - q_1| + |p_2 - q_2|$$

And the generalized formula for an n-D space is given as:

$$D_m = \sum_{i=1}^{n} |p_i - q_i|$$

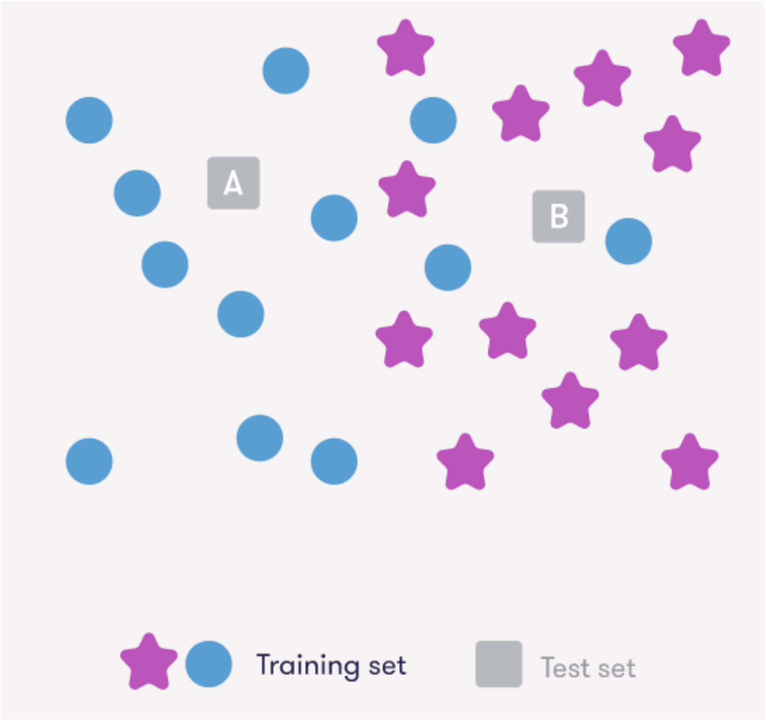


https://www.analyticsvidhya.com/blog/2020/02/4-types-of-distance-metrics-in-machine-learning/

Task 1

If the classification is done based on the color of the single nearest neighbor:

a) A is purple, B is blue
b) A is blue, B is purple
c) Both are blue
d) Both are purple



Task 2

If the classification is done based on the color of the **five** nearest neighbour:

a) A is purple, B is blue
b) A is blue, B is purple
c) Both are blue
d) Both are purple

Task 3

In general, not related to the image above, what happens if you determine the class of a test item by the class of K neighbors, where K is equal to the number of data points you have in your training set?

a) It will work great, because you compare to more data points, thus you get more precise result.
b) It will classify everything to the most prominent class.
c) It won’t work because the algorithm only works for small number of neighbors.

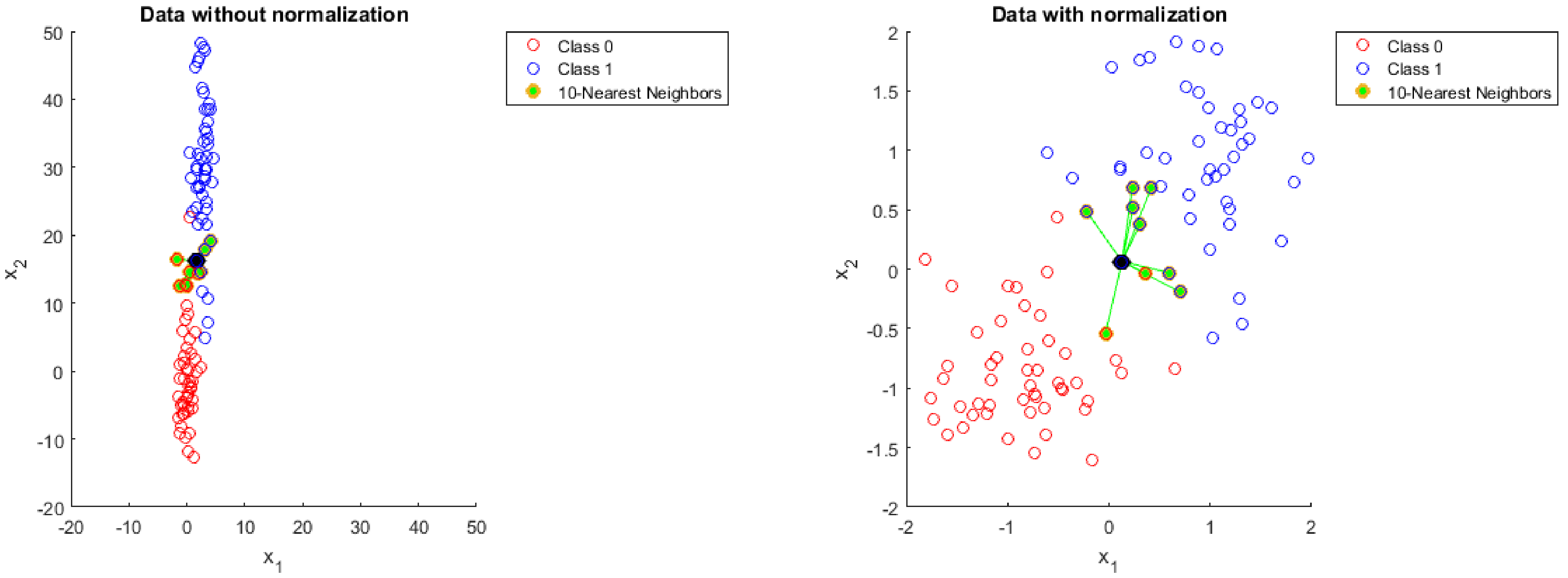
Data normalization and standardization

- However, when we deal with features as a numbers, then the normalization or standardization must be used.

- Suppose we have dataset with two features. One feature is in the range between 0 and 1, while the second feature has values in the range from -1000 to 10000.

- When taking the Euclidean distance between pairs of "examples", the values of the feature dimensions that range between 0 and 1 may become uninformative and the algorithm would essentially rely on the single dimension whose values are substantially larger.

Example - normalization



Feature scaling

- Generally, the problem with normalization or standardization is called as feature scaling. Why to use feature scaling?
    - Scaling guarantees that all features are on a comparable scale and have comparable ranges – this is important for number of methods (k-NN, PCA etc.). Without scaling, bigger scale features could dominate the learning, producing skewed outcomes.
    - Algorithm performance improvement: When the features are scaled, several machine learning methods, including gradient descent-based algorithms, distance-based algorithms (such k-nearest neighbours), and support vector machines, perform better or converge more quickly.
    - Preventing numerical instability: for example, distance calculations or matrix operations, where having features with radically differing scales can result in numerical overflow or underflow problems.

Feature scaling

- Min-Max normalization: $$X_{\text{scaled}} = \frac{X_i - X_{\text{min}}}{X_{\text{max}} - X_{\text{min}}}$$

- Standardization: $$X_{\text{scaled}} = \frac{X_i - X_{\text{mean}}}{\sigma}$$

The scaled version data is often called as Z-score.

- Robust scaling: $$X_{\text{scaled}} = \frac{X_i - X_{\text{median}}}{IQR}$$

IQR – inter-quantile range

$$X_{\text{scaled}} = \frac{X_i - X_{\min}}{X_{\max} - X_{\min}}$$

$$X_{\text{scaled}} = \frac{X_i - X_{\text{mean}}}{\sigma}$$

Normalization	Standardization
Minimum and maximum value of features are used for scaling	Mean and standard deviation is used for scaling.
It is used when features are of different scales.	It is used when we want to ensure zero mean and unit standard deviation.
Scales values between [0, 1] or [-1, 1].	It is not bounded to a certain range.
It is really affected by outliers.	It is much less affected by outliers.
Scikit-Learn provides a transformer called `MinMaxScaler` for Normalization.	Scikit-Learn provides a transformer called `StandardScaler` for standardization.
This transformation squishes the n-dimensional data into an n-dimensional unit hypercube.	It translates the data to the mean vector of original data to the origin and squishes or expands.
It is useful when we don't know about the distribution	It is useful when the feature distribution is Normal or Gaussian.
It is a often called as Scaling Normalization	It is a often called as Z-Score Normalization.

Hamming Distance for strings

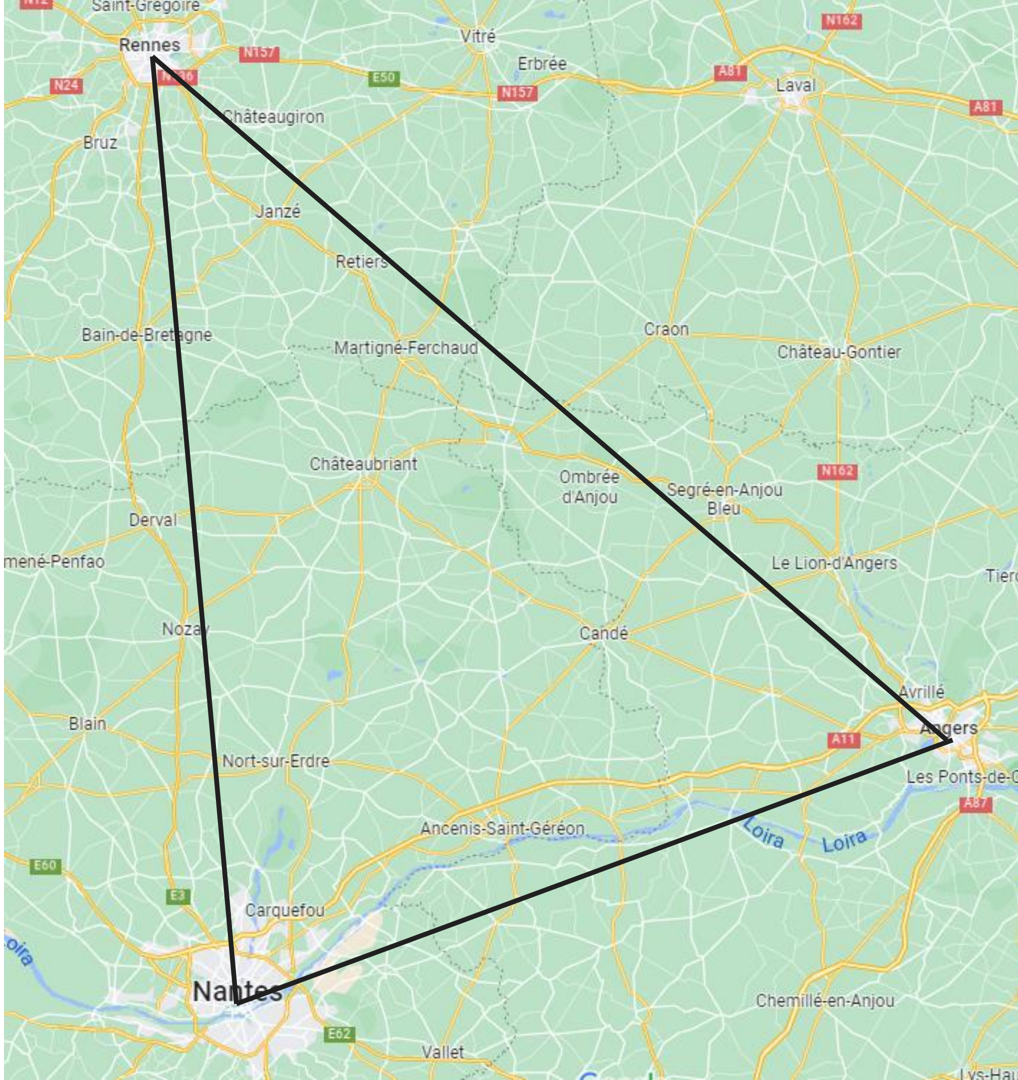
Hamming Distance measures the similarity between two strings of the same length. The Hamming Distance between two strings of the same length is the number of positions at which the corresponding characters are different.

Consider two strings:
A n g e r s
N a n t e s

Hamming distance is 5

Consider two strings:
R e n n e s
N a n t e s

Hamming distance is 3



Shopping prediction – recommendation system

Let’s build a simple recommendation system for an online shopping application where the users' purchase history is recorded and used to predict which products the user is likely to buy next.

We have data from six users. For each user, we have recorded their recent shopping history of four items and the item they bought after buying these four items:

The most recent purchase is the one in the rightmost column, so for example, after buying a t-shirt, flip flops, sunglasses, and Moby Dick (novel), Ville bought sunscreen.

Our hypothesis is that after buying similar items, other users are also likely to buy sunscreen.

To apply the nearest neighbor method, we need to define what we mean by nearest. This can be done in many different ways, some of which work better than others. Let’s use the shopping history to define the similarity (“nearness”) by counting how many of the items have been purchased by both users.

User	**Shopping History**				**Purchase**
Sanni	boxing gloves	Moby Dick (novel)	headphones	sunglasses	coffee beans
Jouni	t-shirt	coffee beans	coffee maker	coffee beans	coffee beans
Janina	sunglasses	sneakers	t-shirt	sneakers	ragg wool socks
Henrik	2001: A Space Odyssey (dvd)	headphones	t-shirt	boxing gloves	flip flops
Ville	t-shirt	flip flops	sunglasses	Moby Dick (novel)	sunscreen
Teemu	Moby Dick (novel)	coffee beans	2001: A Space Odyssey (dvd)	headphones	coffee beans

Our task is to predict the next purchase of customer Travis who has bought the following products:

User	Shopping History				Purchase
Travis	green tea	t-shirt	sunglasses	flip flops	?

Our tasks

1. Calculate the similarity ("nearness") of Travis relative to the six users in the training data.
2. Having calculated the similarities, identify the user who is most similar to Travis by selecting the largest of the calculated similarities.
3. Predict what Travis is likely to purchase next by looking at the most recent purchase of the most similar user from the previous step.

User	Shopping History				Purchase
Sanni	boxing gloves	Moby Dick (novel)	headphones	sunglasses	coffee beans
Jouni	t-shirt	coffee beans	coffee maker	coffee beans	coffee beans
Janina	sunglasses	sneakers	t-shirt	sneakers	ragg wool socks
Henrik	2001: A Space Odyssey (dvd)	headphones	t-shirt	boxing gloves	flip flops
Ville	t-shirt	flip flops	sunglasses	Moby Dick (novel)	sunscreen
Teemu	Moby Dick (novel)	coffee beans	2001: A Space Odyssey (dvd)	headphones	coffee beans

User	Shopping History				Purchase
Travis	green tea	t-shirt	sunglasses	flip flops	?

Shopping recommendation system - conclusion

- In the example, we only had six users' data and our prediction was probably very unreliable.
- However, online shopping sites often have millions of users, and the amount of data they produce is large.
- In many cases, there are many users whose past behavior is very similar to yours, and whose purchase history gives a pretty good indication of your interests.

Filter or social bubbles

- Basically, the recommendation systems of songs (Spotify), videos (Youtube) or contributions (Facebook, Instagram) work very similar.
- So in social media, if the users are likely to click or like something, then it may lead to *filter bubbles* where the users only see content that is in line with their own values and views.

You should be able to:

- Describe principle of Turing test.
- Give some definition/description of *Machine learning* discipline.
- Define differences and similarities between classification and regression.
- Draw and describe typical classification pipeline.
- Define - multilabel and multiclass classification, supervised and un-supervised. learning, describe reinforcement learning.
- Describe K-Nearest Neighbourhood classification, pros/cons, definition of distances.
- Data scaling - normalization, standardization, properties, differences.